Shiven Mahajan

Backend & Full-Stack Software Engineer

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Experience

Sutherland Global Services *Sep 2025 – Present*
Associate App Developer — eSeal Supply Chain Product Team *Hyderabad, India*

- Cut query time from **4s to 1s** and bulk-insert time from **2s to 0.5s** on **20,000+ row** datasets by eliminating N+1 queries and batching writes across an expanded **50+ table** PostgreSQL/SQLite schema.
- Reduced execution time by **71–89%** across product, packaging-level, component, and attribute operations in the product replacement and packaging workflows (e.g., **1.4s to 290ms**).
- Built **ASP.NET Core Web APIs** hosted on IIS and integrated **20+ REST APIs** (authentication, settings, product sync, app updates) into a C#/.NET WPF desktop app (MVVM, service-layer abstractions); configured LAN hosting for scanner-to-desktop communication in warehouses.
- Built from scratch the product, production, and purchase-order workflows: IoT code allocation, multi-level packing (article/carton/master-carton), batch operations, and QR label printing via PRN files with embedded IoT codes.
- Automated warehouse tracking by integrating UF3S/UHF RFID scanners (Bluetooth, TCP, USB) directly into the eSeal app, replacing manual data migration and streamlining high-volume operations across **100,000+ tracking codes**.

Quadrafort Technologies *May 2024 – Jun 2024*
Web & Business Intern *Remote*

- Designed analytical dashboard screens in **Figma** and mapped cross-functional business workflows to guide the team's implementation.

Indian Institute of Science (IISc) *Dec 2023 – Jan 2024*
Research Intern — SPWiCOM Lab, 6G Systems *Bengaluru, India*

- Simulated OTFS and OFDM communication systems in MATLAB and Python across multiple channel environments, running **10,000+ transmission iterations**.

Projects

The Scout: Intelligent Football Transfer Engine *Java, Spring Boot, Python, PyTorch, PostgreSQL*

- Built a modular recommendation platform (Spring Boot API, FastAPI ML inference, PostgreSQL/pgvector retrieval, Redis cache) so each layer can scale independently.
- Trained a Two-Tower model on **1,050 players and 49 clubs** (42 player and 18 club features) to return top-3 transfer recommendations under Total Cost of Ownership constraints; Redis LRU caching cut repeat-request latency from **450ms to 25ms**; the Two-Tower design lets club embeddings be precomputed and served through vector search.
- Achieved an average **Recall@10 of ∼0.4** across held-out clubs, evaluated on historical transfers.

3ngage Engagement Platform *TypeScript, NestJS, Next.js, PostgreSQL*

- Architected a multi-tenant SaaS platform (lightweight browser SDK, NestJS backend, Next.js dashboard, PostgreSQL) for event-driven campaigns, successfully deploying and testing the interactive client-side avatar across **9-page** web environments.
- Designed extensible event, trigger, and rule engines with **tenant isolation and RBAC**, engineering a real-time analytics dashboard capable of processing and visualizing **12 events per minute**.

FPGA Cryptographic Accelerator *VHDL, Vivado, FPGA*

- Designed AES-128 and DES cores in synthesizable VHDL, verified against known-answer vectors, and synthesized in Vivado at **205 MHz (AES)** and **452 MHz (DES)**.

Technical Competencies

Languages: C#, Java, TypeScript, Python, SQL

Frameworks: ASP.NET Core, .NET, WPF, Spring Boot, NestJS, Next.js, React, FastAPI

Backend/DB: REST APIs, PostgreSQL, pgvector, Redis, SQLite

Architecture: MVVM, Event-Driven Architecture, Rule Engines, Multi-Tenancy

ML/Data: PyTorch, Vector Search, Embeddings

Tools: Git, Azure DevOps, Docker, CI/CD, MSI Deployment

Education & Achievements

National Institute of Technology, Srinagar *B.Tech, Electronics & Communication Engineering*
CGPA: 7.48/10 *2021 – Aug 2025*

- GATE EC 2025 Qualifier — Silver Medalist, National Mathematics Olympiad

Certifications: Anthropic Certified: Claude with the Anthropic API, Claude Code Foundations